MarketMind AI - ML Models Used in Milestone 2

What each model powers, why it was selected, and how it supports a real business decision.

1. Selected models and their purpose

Customer segmentation

Selected: K-Means (Hierarchical used for comparison)

Used for: Grouping customers from RFM, purchasing behaviour, returns and engagement features.

Why it fits: There is no correct segment label in the source data, so this is an unsupervised problem. K-Means is scalable, repeatable and produces clear customer groups.

Business help: Creates Champions, Needs Attention and Hibernating groups for retention, offers and prioritisation.

5,878 customers	3 segments	Silhouette 0.364

Business revenue forecasting

Selected: Weekday-aware Linear Trend

Used for: Predicting daily net revenue in INR for 7, 14 and 30 days.

Why it fits: On the current short Amazon history, a simple trend plus weekday pattern had the lowest validation MAE among Seasonal Naive, Prophet, XGBoost and Random Forest.

Business help: Gives the Business Owner a forward revenue view, growth direction and prediction range.

MAE 77,902	RMSE 107,282	25% lower MAE

Sales Executive personal forecast

Selected: Seasonal Naive

Used for: Predicting the signed-in Sales Executive's own daily sales for 7, 14 and 30 days.

Why it fits: The personal dataset is small, and the recent weekly pattern beat the more complex models on the project's MAE-first selection rule.

Business help: Provides a realistic personal target view without exposing another seller's records.

207 source rows	MAE 748	Seller-scoped

Store and product demand forecasting

Selected: XGBoost Regressor

Used for: Predicting daily store-product demand for 7, 14 and 30 days.

Why it fits: Demand depends on non-linear combinations of lag history, calendar, discounts, activity, stock and weather. XGBoost captured these relationships best on chronological evaluation data.

Business help: Helps Store Managers anticipate demand, review stock risk and plan replenishment.

250 series	MAE 2.152	R2 0.918

Important: Administrator model monitoring is not another prediction model. It displays model versions, metrics, status and forecast jobs across the secured APIs.

How MarketMind chooses and uses a model

The system does not select a complex model just because it sounds advanced. Each use case is evaluated against its own data and business target.

2. Repeatable model-selection workflow

Step	Stage	Description
1	Prepare data	Clean records and build reliable features
2	Split correctly	Chronological validation for forecasts
3	Run baseline	Seasonal Naive gives a simple benchmark
4	Compare candidates	Evaluate models on the same holdout data
5	Select + version	Lowest MAE, then RMSE; save metrics
6	Serve securely	Import to DB, API scope, dashboard

3. How to explain the evaluation metrics

Metric	Simple meaning	How to read it in MarketMind
MAE	Average size of the prediction error	Lower is better; this is the primary forecast-selection metric
RMSE	Penalises larger mistakes more strongly	Lower is better; used as the tie-breaker after MAE
R2	How much variation the model explains	Closer to 1 is better; negative revenue R2 means the current history is not production-ready
Silhouette	How separate and compact customer clusters are	Closer to 1 is better; 0.364 gives useful but improvable customer separation

What to say during the presentation

We did not force one algorithm onto every feature. Customer segmentation has no target label, so we use clustering. Revenue and personal sales are time-series problems, while product demand contains many interacting business features. We compare suitable candidates on validation data, select the best result for that use case, version it, import its predictions and expose them through role-protected APIs.

Honest project limits

Revenue history is short, so its negative R2 must be presented as a functional prototype rather than production accuracy. The Parquet demand target is kept as source_unit until the provider confirms whether sale_amount means quantity, value or an index.